

PRANVEER SINGH INSTITUTE OF TECHNOLOGY KANPUR

Odd Semester

Session 2025-26

CT-I

B. Tech.- 5th Semester

Web Technology (BCS502/BCS502H)

Course Outcome	
CO	
CO1	Able to define [L1: Remember] Web Development Strategies, Internet Tools and the protocols that govern the web, Internet Services, basic HTML tags and CSS.
CO2	Able to describe [L2: Understand] web development strategies, architecture of the client-server computing, concepts of JavaScript, XML, Java Networking APIs, Enterprise Java Beans, AJAX, Servlets and JSP.
CO3	Able to demonstrate [L3: Apply] the concepts of HTML, CSS, JavaScript, Node.js, Express Framework, Servlet, JSP, Java Networking concepts such as Internet addressing & TCP/IP Sockets, and database operation using MongoDB.
CO4	Able to examine [L4: Analyze] the different kinds of architecture of web, XML Parsers, different types of CSS, JavaScript and various advance JavaScript concepts for server-side programming, RESTful APIs and Beans.
CO5	Able to compare [L5: Evaluate] dynamic and client & server-side interactive web applications.

Time: 1.5 Hrs.

M. M. 20

Section A

(1X5 = 5 Marks)

Q1. Attempt all questions:

- State the history of web with a brief about each era.
- List the different attributes and tags of Frames in HTML citing an example for the same.
- State any two web-based protocols used in communication.
- Describe any two internet services used to exchange information.
- Explain different ways of object creation in JavaScript.

CO1
CO1
CO1
CO2
CO2

Section B

(2.5X4 = 10 Marks)

Q2. Attempt all questions:

- Explain any two web development strategies to be followed by a web developer.
- OR
- Differentiate between DOM and SAX processors of XML.
- Explain about functions in JavaScript, stating different ways of creating a function.
- OR
- Explain about the client-server TCP/IP sockets by citing an example showing communication between them.
- Express AJAX in detail and debate on the advantages and disadvantages of it.
- OR
- Explain about different pop-up boxes available in JavaScript citing example of each with syntax.

CO2
CO2
CO2
CO2
CO2

- d i) Apply the table concept of HTML in designing the following table having solid border as 5px, using col spanning and row spanning: CO3

A	B		C
D	E	F	G
	H	I	

OR

- ii) Illustrate javascript concept to create a web page with a registration form that validates user input using JavaScript. CO3
- Name (required, minimum 2 characters)
 - Email (required, valid email format)
 - Password (required, minimum 6 characters)
 - Age (must be between 18 and 100)

Section C

Q3.

(5X1 = 5 Marks)

- i) Apply HTML and styling concept to design the following: CO3

- Fruits
 - Apple
 - Banana
 - Grapes
 - c. Red Grapes
 - d. Green Grapes
- Vegetables
 - 3. Carrot
 - 2. Broccoli
 - 1. Capsicum

The whole list background is yellow. Red Grapes is in red color; Green Grapes are in green color.

OR

- ii) Illustrate XML and XSD concept by creating an XML document for a library book catalog and an XSD schema with the following restrictions: CO3
- Each book must have: ISBN (unique), title, author, genre, and publication year
 - ISBN must follow the pattern: 3 digits - 2 digits - 4 digits (e.g., "123-45-6789")
 - Title and author must be between 1 and 50 characters
 - Genre must be one of: DAA, Compiler, Web Technology, DBMS
 - Publication year must be between 2000 and current year